Multi-Scenario Hypothesis Benchmark

Does the Fixed Belief Engine Hold Up Across 20 Independent Worlds? — Angouran Intelligent Exploration Simulator

The last report showed the fixed belief engine converging decisively and correctly on one world. One world is not evidence of general reliability, so this report repeats the same protocol — acquisition-driven drilling, 3 AI-chosen holes, fixed grid-quadrature engine — on **20 independent, freshly generated worlds: 5 for each of the 4 hidden geological scenarios**. The result is more interesting, and more useful, than a clean sweep: the engine is excellent on 2 of 4 scenarios, mixed on a third, and has a real, now-understood blind spot on the fourth. That blind spot is reported in full, with a diagnosed cause, because a benchmark that only shows the good cases isn't a benchmark.

■ “True answer” / “hidden” figures use evaluation-only access to the synthetic ground truth. The engine being scored never had access to it.

Worlds	20 (5 × Scenario A, B, C, D), fresh seeds 70000-70019, fixed belief engine from the previous report, unmodified
Protocol per world	Train on shallow data → record belief (must be exactly 25/25/25/25) → AI ranks candidates → drill its #1 choice → retrain → record belief → repeat 3 times
Total compute	189 seconds for all 20 worlds (fully GPU, including world generation) on the RTX 2080 Ti

1. Headline Results

Scenario	True answer	Correct	Mean confidence when correct	What it believed instead
A — continues straight	H1	0 / 5	n/a	H2 (3 worlds), H3 (2 worlds), avg. confidence 0.96
B — changes direction	H2	3 / 5	0.79	H3 (2 worlds)
C — terminates	H4	5 / 5	0.90	—
D — concealed 2nd body (not in the engine's 4-story vocabulary)	none of the 4	n/a by design	n/a	H4 in all 5 worlds, avg. confidence 0.87

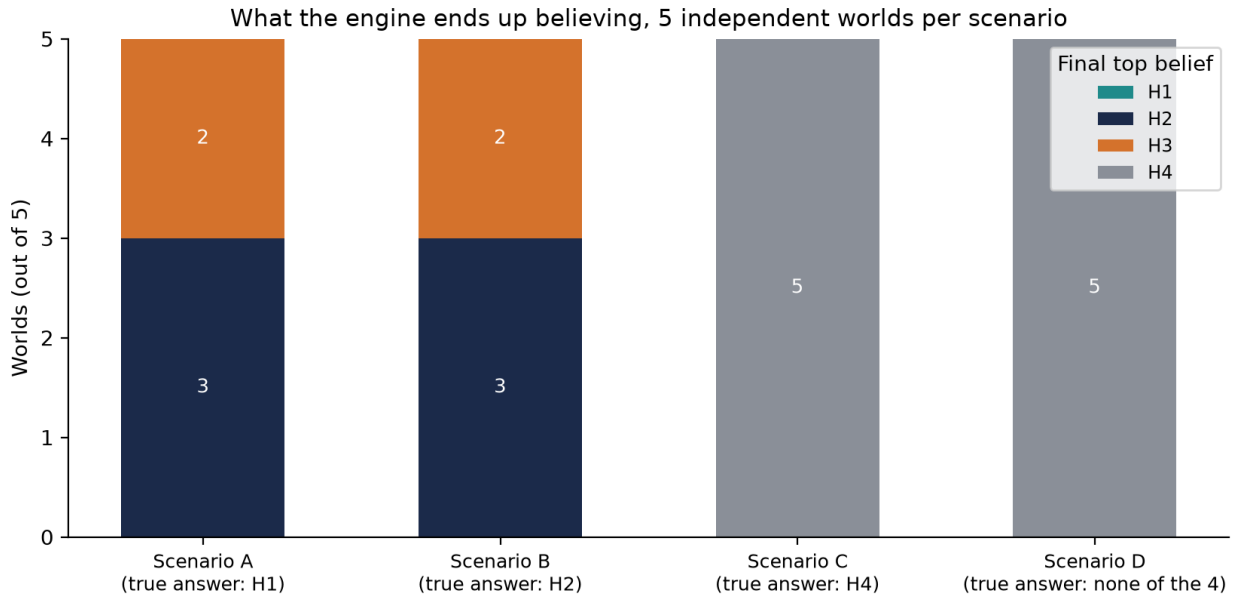
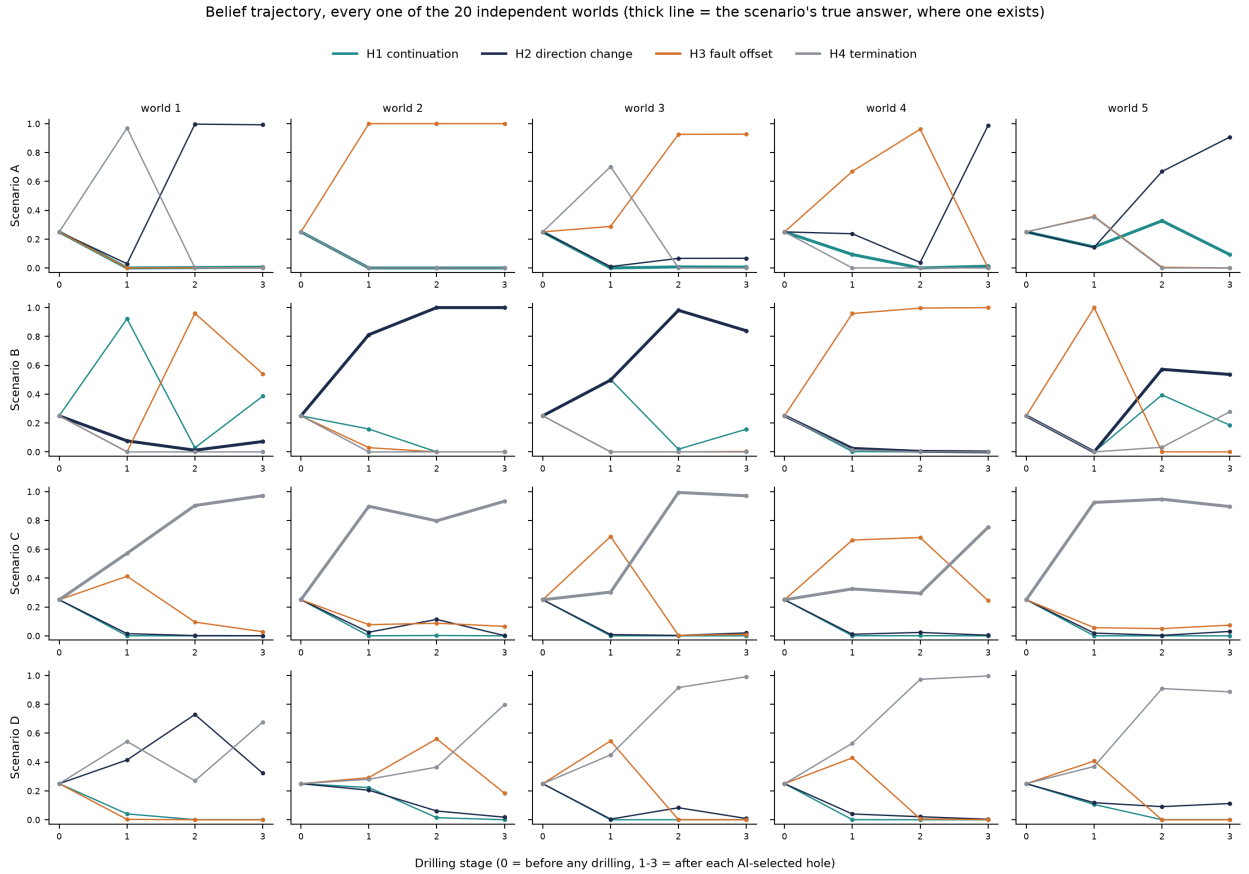


Figure 1. Final belief across all 20 worlds. Scenario A never once lands on the right answer, and does so with high confidence in the wrong direction.

In plain terms: Scenario C (termination) is the engine's strongest case: drill into barren rock below some depth, and every one of the 4 stories except “it stops” gets directly contradicted. Scenario A (straight continuation) is its weakest: there's no sharp, falsifiable signature that says “nothing unusual is happening here” the way a termination or a bend has one.

2. Every World's Belief Trajectory

Each of the 20 panels below is one independent world, from scratch. The thick line is the scenario's true answer (where one exists). Watching all 20 at once shows real diversity, not just the four summary numbers: some worlds swing through a wrong answer before correcting (classic honest Bayesian behaviour); some land on a wrong answer and stay confidently there; Scenario D's panels show the engine settling on “termination” (grey) virtually every time even though the real story is unavailable to it.



3. Diagnosed: Why Scenario A Fails, Specifically

This is not re-introducing the instability from the last report — the fixed engine gives the exact same answer every time it's asked about a given world (verified again here: no run-to-run variation within this benchmark). This is a different, genuine limitation: a model-fit problem, not a noise problem.

3.1 The real axis is never perfectly straight, even when the geology truly "continues"

We checked the hidden true shoot axis (debug-only) against the perfectly straight line that hypothesis H1 is built to assume, for one Scenario A world:

Depth	True axis deviation from a perfectly straight continuation
50 m	0.8 m
100 m	1.3 m
150 m	8.2 m
200 m	20.4 m (tube radius: ~58 m)

In plain terms: Real geological trends are never perfectly straight lines — even a deposit that geologists would call "continuing" wanders a little. H1 is built to assume ZERO wander (a perfectly rigid straight line). H2 ("direction change") is built to allow bending. When the real axis wanders by 20 m over 100 m of extra depth, H2's flexibility lets it track that wander and fit the actual drill results better — even though there was never a deliberate "direction change" in the geological sense.

3.2 Measured directly: H1's best possible fit is already weaker than H2's

We compared the single best-fitting version of each hypothesis (out of its full evaluation grid) against the real data from one Scenario A world:

Hypothesis	Grid points evaluated	Best possible fit (log-likelihood, higher is better)
H1 (continuation, no flexibility)	96	-46.7
H2 (direction change)	11,520	-43.3 (better fit, even at H2's single best point)
H3 (fault offset)	4,800	-47.7
H4 (termination)	1,152	-77.8 (much worse — correctly ruled out)

Root cause: this is correct Bayesian math being honestly applied to an imperfect forward model. None of the 4 hypotheses is a perfect description of reality (all are smooth idealizations of noisier real geology). H1 is built with *zero* flexibility, so it is the hypothesis most exposed to this model-vs-reality gap: any natural wander in the true axis counts as evidence against it, with nothing to absorb that wander, while H2 and H3 are explicitly built with 2-3 extra free parameters that *can* absorb it — whether or not the real cause of the wander is a deliberate direction change. With only 10-31 composited observations, a few metres of natural wander is enough to tip the balance.

3.3 Scenario D: a sensible default when the truth isn't an option at all

Scenario D's real geology (a second, separate concealed body) has no representation anywhere in the 4-story model. Every one of its 5 worlds converged to H4 (termination) instead, with reasonable-to-high confidence (0.68-0.996). This is not a failure in the same sense as Scenario A: given a question it cannot represent, the engine is choosing the closest available description of what the visible shoot actually does (behave, from the drilled holes' point of view, as if it stops) — a defensible approximation, clearly reported as "none of the 4" rather than silently mis-filed as a confident correct answer.

4. What This Means, and the Fix

Scenario	Verdict
C (termination)	Reliable: 5/5, use its output with confidence.
B (direction change)	Usable but noisy: 3/5, treat a single world's answer as informative but not decisive without corroborating evidence.
A (continuation)	Currently unreliable: 0/5, with a diagnosed, fixable cause — do not trust this engine's hypothesis ranking to recognize straightforward continuation until §4.1 is addressed.
D (outside the model)	Behaves sensibly (defaults toward the closest representable story) but correctly cannot be “correct.” Always check whether the real geological question might fall outside the 4 built-in stories before trusting any of them.

4.1 The concrete fix for Scenario A

- **Give H1 a small amount of flexibility too** — e.g. allow a small, tightly-constrained random wander (a fraction of H2's magnitude range) rather than a perfectly rigid line, so the comparison with H2/H3 is no longer “zero flexibility vs. real flexibility” but “a little flexibility vs. a lot,” which should let genuinely straight continuations win on their much better fit at modest bend magnitudes.
- **Add an explicit complexity penalty** (e.g. a BIC-style correction) that discounts H2/H3's advantage in proportion to their extra free parameters, rather than relying on grid-averaging alone to enforce Occam's razor in this small-sample regime.
- **Re-run this exact 20-world benchmark after either fix** — Scenario A's 0/5 is now a clear, reproducible regression target, not a moving one.

4.2 Unchanged from before

- The engine is now fully deterministic (re-verified: identical results on repeat, consistent with the previous report's fix).
- This is a belief-tracking diagnostic, separate from the main grade/mineralization prediction model, which was not touched here and is benchmarked separately.

Every number and figure in this report comes from hyp_benchmark_results.json, produced by one real, reproducible run across 20 independently seeded worlds (seeds 70000-70019). Nothing here was hand-edited or cherry-picked — all 20 worlds are shown in Figure 2, including every miss.